AuctionEngine

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Context first. Pressure second. Execution last.

About This Guide

This is the **product reference** for the Flow.42 AuctionEngine. It covers installation, every on-chart element, every property, the **State Panel** read line by line, the **State Change Markers**, and the optional **AI Analyst**.

The AuctionEngine is a **context tool**, not a trade signal generator. It reads the market the way an experienced order-flow trader does — building a session volume profile, judging where price sits relative to value, and naming what the auction is doing right now. It tells you *what kind of market you are in* so you, and the rest of the Flow.42 Trading Suite, can adapt.

> **For the concepts behind the auction** — volume profile theory, market auction logic, the full meaning of each state, and trade scenarios — see the separate **Flow.42 AuctionEngine Course**. This guide stays focused on *the product*: what each setting does and what each pixel on the chart means.

1. Installation & Requirements

The AuctionEngine ships as a set of NinjaScript files. They all belong to the same source tree and must sit together in your NinjaTrader Documents\NinjaTrader

8\bin\Custom\Indicators folder and compile as one:

File	Role
Flow42AuctionTypes.cs	Shared public types — the required foundation.
Flow42AuctionCore.cs	Profile, classifier and public API — the engine logic.
Flow42AuctionHub.cs	Optional shared-settings hub.
Flow42AuctionAnalyst.cs	Optional AI Analyst client (backend call + result window).
Flow42AuctionEngine.cs	The chart indicator itself.

The engine **will not compile** without Types and Core. The Analyst file is only needed if you intend to use the AI Analyst button; the Hub is only needed for shared settings across charts.

Volume data matters

The quality of the profile depends entirely on the volume source (see *Volume Profile Model* in the property reference). For an accurate Volume-at-Price profile, use one of:

- **Tick Replay** — requires Tick Replay enabled on the data series. Builds the profile from actual trades.
- **Volumetric Bars** — uses an Order Flow Volumetric series.

The **bar-distributed approximation** is a fallback for feeds without true Volume-at-Price. It works, but the profile is an estimate, not a measurement.

Adding it to a chart

Add **Flow.42 AuctionEngine** to a chart like any other indicator. Core and Types are libraries — you never add them to the chart directly. The Hub is added only if you want shared settings (see *Settings Source*).

> **License.** The AuctionEngine is a licensed Flow.42 product. The optional AI Analyst additionally checks your saved Flow42 license id against an active Flow.42 Trading

> Suite subscription. Enter your license in NinjaTrader under the indicator's **Product Information** section.

2. Quick Concept Overview

The engine builds a session volume profile, classifies the current market behavior as one of **13 auction states**, and draws value-reference lines. Five terms appear throughout — learn these and the rest follows:

Term	Meaning
Profile / Value	The price region where most time and volume traded — where the market saw "value."
VPOC	Volume Point of Control — the single price with the most volume. The magnet of the session.
VA / VAH / VAL	Value Area (default ~70% of volume). VAH = upper edge, VAL = lower edge.
developing (d...)	Today's live session, still forming: dVPOC, dVAH, dVAL.
prior (p...)	The previous, completed session — a fixed reference: pVPOC, pVAH, pVAL.

Two more nodes appear on the chart: **HVN** (High-Volume Node — a magnet inside the prior profile) and **LVN** (Low-Volume Node — a fast-through decision zone). Full theory lives in the Course.

3. On-Chart Elements

The engine renders several independent, draggable elements:

- **Reference lines** — horizontal lines for prior & developing VPOC/VAH/VAL, prior range (high/low), session open, and HVN/LVN zones. Optional labels sit at the chart edge.
- **State Panel** — the draggable text box showing the current auction state, confidence, location, profile values and (optionally) the detail lines.

- **Summary box** — a separate compact box with the current profile values.
- **Overview table** — a separate table summarizing the last N completed sessions.
- **State Change Markers** — vertical dashed lines + compact labels where the classified state changes (e.g. “IMB UP”, “FAIL DOWN”).
- **Chart buttons** — a horizontal row of clickable buttons: the **Panels** button, the **Auction Engine** toggle, and (optional) **Analyze**.

Each box — panel, summary box, overview table, button row — can be dragged independently and positioned via its anchor + offset properties.

The Panels button *(new — 2026-07-09)*

The leftmost chart button is split into three independently clickable segments:

State · Summary · Table

Each segment shows or hides one panel: the **State Panel**, the **Summary box** and the **Overview table**. **Green means visible, gray means hidden** — the same color language as the Auction Engine button, which is why no ON/OFF text is needed.

- **Click a segment label** to toggle that panel. Clicking the padding or a separator does nothing.
- **Drag anywhere on the button** to reposition the whole button row.
- The engine keeps running either way. Reference lines and state-change markers stay visible even with all three panels hidden.
- The toggles are a **temporary view, not a saved setting**. On chart reload each panel returns to the state configured in its property (*Show Panel*, *Show Summary Box*, *Show Overview Table*). Use the properties for the layout you always want; use the button to take a quick look.

> **Overview table.** Its rows come from the engine’s rolling daily-profile cache, not from the property, so switching the table on mid-session shows real data immediately — you do not have to wait for sessions to accumulate.

4. The State Panel — Line by Line

The State Panel is the small text box on the chart. It reads the volume profile per session and tells you, in a few lines, **what the market is doing right now**. It is a context and orientation tool — never a buy/sell signal.

> **Hover = time travel.** Move the mouse over a past candle and the panel rolls back to show the state **at that exact moment**, with the Session date adjusting to that bar's day. Without hovering, it shows the most recent state.

Head & core read

Line	Example	What it tells you
(Title)	Flow.42 AuctionEngine	Just the header.
Session	Session: RTH (26.06.)	Which profile session is evaluated (RTH / ETH / Asian / Custom / ChartTradingHours) and the date of the hovered or current bar.
State	State: Imbalance Up	The most important line. The detected auction state (full list in §5). Text color shows direction: green = bullish, red = bearish, gold = caution, gray = neutral.
Conf	Conf: 72 / 100	Confidence — how clear/strong the read is (0–100). High = clean picture, low = shaky.
Loc	Loc: Above prior value	Location — where price sits relative to the prior session's value area (see §6).

Profile figures

Line	Example	What it tells you
Events / Calc / Vol	Events: 1234 Calc: Tick Vol: 85213	Diagnostics: how many profile events were processed, what last triggered the calculation, and the session

Line	Example	What it tells you
		volume so far. Technical control.
dVPOC / dVA	dVPOC: 5012.50 dVA: 5008.00 - 5020.00	Today's (developing) session: VPOC and value area (VAL–VAH). The key lines of the live session.
pVPOC / pVA	pVPOC: 4995.00 pVA: 4988.00 - 5004.00	Previous (prior) session: VPOC and value area — the central reference lines for today.
InValue / Above / Below	InValue Bars: 64% AboveVol: 22% BelowVol: 14%	Acceptance stats vs. prior value: share of bars inside prior value, share of volume above prior VAH, share below prior VAL. Shows whether the market accepts the old value (stays in) or rejects it (breaks out).
(Substitute line)	Prior: load more history	Appears instead of the prior lines when no prior session is loaded yet → load more chart history.

Detail block (only when Show Detail is on)

Line	Example	What it tells you
Invalid	Invalid: below 5008.00	Invalidation — the level/condition at which the current read is "broken." If violated, the state no longer holds. The engine's stop-thinking.
Target	Target: 5020.00	Where the market could go if the read plays out (e.g. back to VPOC, to a value edge).
Reason	Reason: accepting value higher	Short reason why the engine assigned this state.
Hint	Hint: watch for pullback into value	A short, action-near observation: what to watch next. A tip, not an order.

5. The Auction States

The State line is the core read. Here are all 13 states in plain terms, plus the directional bias the engine attaches to each. Color follows bias: green bullish, red bearish, gold caution, gray neutral.

State	Bias	Meaning
Initializing	neutral	Too little data — the profile is still building.
Open Inside Prior Value	neutral	Opened inside prior value → market accepts the old value (often a range day).
Open Above Prior Value	caution	Opened above prior value → strength, but must confirm.
Open Below Prior Value	caution	Opened below prior value → weakness, must confirm.
Balance / Rotational	neutral	Two-sided, rotating market — a range, no clear trend.
Mid-Value / No Edge	caution	Price mid-value, no advantage for either side — a poor place to trade.
Higher Value Acceptance	bullish	Market accepts value higher → upside continuation.
Lower Value Acceptance	bearish	Market accepts value lower → downside continuation.
Failed Upper Auction	bearish	Attempt to auction higher failed → rejection up top → turn down expected.
Failed Lower Auction	bullish	Attempt to auction lower failed → rejection at the bottom → turn up expected.
Imbalance Up	bullish	Strong one-sided imbalance up → buy pressure / drive higher.
Imbalance Down	bearish	Strong one-sided imbalance down → sell pressure / drive lower.
Transition / Unclear	caution	State is shifting, the picture is ambiguous — wait.

> **Mnemonic.** *"Failed" means a turn* — a failed auction in one direction implies a move in the **other**. *"Acceptance" and "Imbalance" mean continuation* in the named direction.

6. The Location Read

The Loc line says **where price sits relative to the prior session's value** — its "location" in the market:

Location	Meaning
Below prior value	Below the prior value area (below VAL).
At prior VAL	At the lower edge of prior value.
Inside prior value	Inside the prior value area.
Near value center	Near the middle (close to VPOC).
At prior VAH	At the upper edge of prior value.
Above prior value	Above the prior value area (above VAH).

The edges (VAL / VAH) and the center (VPOC) are where reactions tend to happen — which is why location matters alongside the state.

7. State Change Markers

What they are. A vertical dashed line with a short label at the exact bar where the auction state changed. They make visible *when* the character of the market shifted — a timeline of turning points.

Important: a marker appears **only on a real state change** into a "notable" state — not on every bar. Early-session and unclear states deliberately get **no** marker.

Marker labels

Label	State	Meaning
IMB UP	Imbalance Up	Shift into strong buy pressure up.
IMB DOWN	Imbalance Down	Shift into strong sell pressure down.
HVA	Higher Value Acceptance	Market begins accepting value higher.
LVA	Lower Value Acceptance	Market begins accepting value lower.
FAIL UP	Failed Upper Auction	Upside auction failed (turn down expected).
FAIL DOWN	Failed Lower Auction	Downside auction failed (turn up expected).
BAL	Balance / Rotational	Shift into a balanced range mode.
MID	Mid-Value / No Edge	Shift into the edgeless middle (no advantage).

Behavior

- Each change is recorded **once** (the same state repeating produces no new marker).
- Markers are held in a **bounded, rolling window** (about **1 day** of look-back) and stay visible **across the session change** — so you also see the prior session's turning points.
- On **instrument change** or shutdown, everything resets.

> **Why the panel color can change *between* two markers.** The panel always shows the **exact state of the bar under your cursor**. Markers, by contrast, are only placed on changes into a *marker-worthy* state — and *Transition / Unclear* and *Mid-Value / No Edge* deliberately get none. So between two visible "FAIL DOWN" markers, individual bars may have briefly flipped to *Transition / Unclear* (gold) without leaving a marker. Hovering those bars shows gold — this is **correct**. The markers give a clean overview; the panel gives the precise truth.

8. Color Quick Reference

The panel text color follows the current state bias:

Color	Bias	States
Green	Bullish	Imbalance Up, Higher Value Acceptance, Failed Lower Auction
Red	Bearish	Imbalance Down, Lower Value Acceptance, Failed Upper Auction
Gold	Caution	Mid-Value / No Edge, Transition / Unclear
Gray	Neutral	The remaining states (Initializing, Open..., Balance)

> Note the inversion for failed auctions: **Failed Upper Auction is bearish (red)**, **Failed Lower Auction is bullish (green)**. A failed push one way implies a move the other way.

9. The AI Analyst

The AI Analyst turns the engine's structured read into a short written analysis. You click a button, the current auction snapshot is sent to the Flow.42 backend, and a narrative explanation comes back in a window on your screen.

> **The division of labor is deliberate.** The engine stays the deterministic classifier — it decides the state, the confidence, the location. The model only **synthesizes and explains** what the engine already concluded. It does not classify, and it does not produce trade recommendations. For analysis and context only.

How it works

- Enable **Show Analyst Button** (off by default). An **Analyze** button appears at the right end of the chart button row.
- Click it. The button reads **Analyzing...** while the request is in flight; further clicks are ignored until it returns.
- A result window opens with three parts: a header (timestamp + instrument), a deterministic **auction gauge**, and the model's written analysis.

The whole thing is **strictly on demand** — it never runs per bar, and never on a render or calculation thread. A click is the only trigger, so the chart is never blocked or slowed.

What gets sent

The snapshot is read from the **same objects the panel renders**, so the model sees exactly what you see. It contains four blocks:

- **Classification now** — state, location, confidence, target, reason, hint, invalidation.
- **The evidence** — prior and developing profiles, the time/volume distribution relative to prior value, and how many bars since the last VAH/VAL test.
- **The day's story** — the state changes of the current session, oldest first.
- **Multi-day context** — up to five most recent completed daily profiles (VPOC/VAH/VAL/high/low).

> **Privacy & licensing.** The client never stores or sends an Anthropic API key. It identifies you with your saved **Flow42 license id** and lets the backend authorize your entitlement — the AI Analyst requires an active **Flow.42 Trading Suite** subscription. If no license id is configured, the window tells you to enter one in the **Product Information** section.

The auction gauge

Above the written analysis sits a fixed-width gauge with a needle running from **BEARISH** through **BALANCED** to **BULLISH**:

> BEARISH ◄——————|——●——————► BULLISH Auction: Higher Value Acceptance | Confidence: 75/100

The needle is computed from the **same classification the panel shows** — never from the model. That keeps it reproducible and always consistent with the engine (an LLM-set needle would vary run to run and could contradict the panel). Each state carries a direction weight, scaled by confidence, to land in [−1, +1]: full deflection for imbalances,

three-quarters for acceptance, two-thirds for failed auctions (deflecting *away* from the failed side), slight for directional opens, and center for balance / mid / transition states.

> The gauge labels are **auction-context language** — BEARISH / BALANCED / BULLISH — deliberately **not** buy/sell signal language. This matches the no-recommendations positioning of the whole tool.

Result window options

The window is dark, scrollable and copyable. Two settings shape it:

- **Analysis Language** — the written analysis can come back in German, English, Brazilian Portuguese, Spanish, Italian, French, or Simplified Chinese. The backend applies the same stable analysis rules across all locales.
- **Window Text Color** — plain **White**, or the classic green-phosphor **Terminal Green** (green on near-black, the CRT look) which many find easier on the eyes.

> **A display failure can never destabilize the chart.** Every failure path — a network error, a backend rejection, no data yet — is reported inside the result window and handled defensively. The header timestamp is stamped at click time on your machine, so you always know exactly when an analysis was made.

10. Property Reference

Properties are grouped exactly as they appear in NinjaTrader. Each lists its default and range where applicable. Properties marked **(Advanced / hidden)** are not shown in the normal grid — they are exposed for calibration via the Hub.

Group 01 — Data Model

Property	Default	What it does
Volume Profile Model	BarDistributedApproximation	Source used to build Volume-at-Price: bar approximation, Tick Replay (last trades), or Order Flow Volumetric. Use Tick Replay or Volumetric for an accurate profile.
Volumetric Instrument Name	(blank)	Optional instrument name for Volumetric mode. Blank = chart instrument (e.g. NQ 06-26).
Volumetric Base Period Type	Minute	Base period type for the secondary Volumetric series. Volumetric mode only.
Volumetric Base Period Value	1 (1–10000)	Base period value for the Volumetric series. Use 1 Minute for the first performance test.
Volumetric Ticks Per Level	1 (1–20)	Ticks grouped into one Volumetric price level. 1 = most precise.

Group 02 — Auction Session

Property	Default	What it does
Settings Source	Local	Use this indicator's own Session/Profile/Classifier settings, or read them from a Flow.42 AuctionHub on the same chart (matched by instrument + interval). Falls back to Local if no matching hub.
Profile Session	RTH	Time window used to build the profile (ChartTradingHours / RTH / ETH / Asian / Custom). RTH recommended as the primary value reference.
Session time windows	RTH 1530–2200, ETH 2200–1530, Asian 100–800	Start/end times (platform time) per session template, in HHmm. Edited via HH:mm pickers. (Advanced / hidden.)

Group 03 — Auction Profile

Property	Default	What it does
Value Area Percent	70.0 (50–95)	Percentage of total profile volume used to calculate VAH and VAL. Common default: 70%.
Smoothing Ticks	3 (1–20)	Neighboring ticks used to smooth HVN/LVN detection. Higher = fewer, less noisy nodes.
Max HVN/LVN Nodes	3 (0–10)	Maximum High/Low Volume Nodes shown from the prior profile.
Max Bar Distribution Ticks	300 (10–2000)	Safety cap for the bar-distributed approximation. Very wide bars are sampled instead of distributing volume across every tick (performance guard).
Buffer Ticks	2 (0–20)	Tick buffer around prior-value references. Avoids classifying tiny one-tick pierces as acceptance/rejection. On small-tick instruments (e.g. CL at 0.01) a low tick count is a very small price distance. (Advanced / hidden.)

Group 04 — Auction State (Advanced)

These eleven values calibrate the classifier. They are **hidden** in the normal grid and exposed via the Hub for per-instrument tuning. Defaults are identical in engine and hub, so switching Local↔FromHub at default values changes nothing.

Property	Default	What it does
Opening Bars	3 (1–20)	Bars treated as the opening phase before stronger classification applies.
Min Acceptance Bars	4 (1–50)	Minimum bars outside prior value before acceptance is confirmed.
Max Failed Auction Bars	5 (1–50)	Max bars outside value for a fast rejection to count as a failed auction.

Property	Default	What it does
Min Acceptance Volume Percent	0.22 (0.01–0.80)	Minimum share of session volume outside prior value to support acceptance.
Near POC Ticks	8 (1–100)	Distance in ticks for classifying price as near the POC.
Balance Min Value Overlap	0.60 (0.10–1.00)	Minimum prior/developing value overlap to classify rotational balance.
Balance Min Inside Bars Percent	0.55 (0.10–1.00)	Minimum share of bars inside prior value to support balance.
POC Stable Width Percent	0.25 (0.05–1.00)	Max POC migration (share of value width) for the POC to count as stable.
Imbalance Migration Width Percent	0.10 (0.01–1.00)	Minimum developing-POC migration (share of value width) to support imbalance.
Range Extension Ticks	8 (1–100)	Minimum range extension beyond the initial balance area used by imbalance detection.
Failed Auction Memory Bars	10 (1–200)	Max bars after a value test during which a Failed Upper/Lower Auction can still be classified.

Group 05 — Lines & Levels

Property	Default	What it does
Show Indicators	true	Master visibility switch for all auction value lines. Off = no lines, labels or summary box.
Show Prior Value	true	Shows prior-session VPOC, VAH and VAL.
Show Prior Session Levels	false	Also draws prior VPOC/VAH/VAL over the prior session's own time span (not just forward).
Show Developing Value	true	Shows developing VPOC, VAH and VAL of the active session.

Property	Default	What it does
Show Prior Range	true	Shows the prior profile high and low as extra references.
Show Current Session Open	true	Shows the opening price of the active profile session.
Show HVN	true	Shows High Volume Node zones from the prior profile.
Show LVN	true	Shows Low Volume Node decision zones from the prior profile.
Show Summary Box	true	Whether the draggable summary box with the current profile values is shown when the chart loads. Toggle it while the chart is open with the **Summary** segment of the Panels button. *(2026-07-09)*
Label Visibility	true	Shows compact labels directly on the price lines (rendered in OnRender).

Group 06 — Lines

Property	Default	What it does
Line Width	2 (1–6)	Base width of the reference lines (VPOC lines may be slightly thicker).
No After Hours Lines	false	On = horizontal lines stop at the configured session end (e.g. RTH end) and do not extend into after-hours.
Show Developing History	false	Draws developing values as a stairstep polyline showing their evolution through the session.
Legacy Label Offset Ticks	2 (-20–20)	Deprecated, no effect. Use Line Label Offset Pixels instead. (Hidden.)

Group 07 — Line Labels

Property	Default	What it does
Line Label Side	Right	Which chart edge labels anchor to (Left/Right).
Line Label Offset Pixels	8 (0–400)	Pixel distance from the chosen edge to the label box.
Text Size	12 (8–32)	Text size for labels on the reference lines.
Background Opacity	75 (0–100)	Opacity of the line-label background.
Background Color	DimGray	Background color behind each line label.
Text Color	White	Label text color when “Use Level Color For Text” is off.
Use Level Color For Text	false	Uses the reference-line color as the label text color instead of the fixed Text Color.

Group 08 — Summary Box

Position: anchor + offset (Right/Top, X 1.0%, Y 2.0% by default), independently draggable.

Property	Default	What it does
Label Box Font Size	13 (8–32)	Text size in the draggable summary box.
Label Box Opacity	50 (0–100)	Background opacity of the summary box.
Label Box Background Color	Black	Background color of the summary box.
Summary Text Color	(Line Colors group)	Text color in the summary box (see Line Colors → Summary Text Color).

Group 09 — Line Colors

All line colors are hidden from the compact grid by default but fully serializable (saved with the chart).

Property	Meaning
Prior VPOC Color	Prior-session VPOC line.
Prior Value Color	Prior VAH & VAL lines.
Developing VPOC Color	Active-session developing VPOC.
Developing Value Color	Developing VAH & VAL.
Prior Range Color	Prior profile high & low.
Session Open Color	Active session open.
HVN Color	Prior-profile High Volume Nodes.
LVN Color	Prior-profile Low Volume Nodes.
Summary Text Color	Text in the summary box.

Group 10 — State Panel

Position: anchor Right/Top, offset X 1.0%, Y 2.0% by default; independently draggable.

Property	Default	What it does
Show Panel	true	Whether the draggable auction context panel is shown when the chart loads. Toggle it while the chart is open with the **State** segment of the Panels button. Note: with this property off, the reference lines and the summary box are hidden as well. *(2026-07-09)*
Show Detail	true	Shows extra diagnostic lines (profile values, volume distribution, Invalidation, Target, Reason, Hint).
Font Size	13 (8–30)	Text size in the panel.
Panel Opacity	45 (0–100)	Background opacity of the panel.

Group 11 — State Panel Colors

Property	Default	What it does
Panel Text Color	LightGray	Default text color in the panel.
Bullish State Color	LimeGreen	Text color for bullish states (Higher Value Acceptance, Imbalance Up, Failed Lower Auction).
Bearish State Color	IndianRed	Text color for bearish states (Lower Value Acceptance, Imbalance Down, Failed Upper Auction).
Neutral State Color	LightGray	Text color for neutral states (Balance / Rotational).
Warning State Color	Goldenrod	Text color for warning states (Mid-Value / No Edge, Transition / Unclear).
Panel Background Color	Black	Background color behind the panel.

Group 12 — State Change Markers

Property	Default	What it does
Show State Change Markers	true	Draws a vertical dashed line + compact label at every bar where the classifier changes to a marker-worthy state.
Line Opacity Percent	60 (0–100)	Opacity of the vertical dashed line. The label itself is always full opacity for readability.
Label Top Offset Pixels	22 (0–200)	Vertical pixel offset of the label below the chart top. Increase if other indicators overlap.

Marker labels: IMB UP, IMB DOWN, HVA, LVA, FAIL UP, FAIL DOWN, BAL, MID. *Transition / Unclear* and *Initializing*/opening states get **no** marker — this is the reason behind the hover behavior in §4.

Group 13 — Overview Table

Property	Default	What it does
Show Overview Table	false	Whether the overview table is shown when the chart loads: a separate draggable table of the last N completed sessions (Date, VPOC, VAH, VAL, High, Low). Toggle it while the chart is open with the **Table** segment of the Panels button. *(2026-07-09)*
Overview Days	5 (3–10)	How many completed sessions the table shows. Days accumulate as sessions complete.

Table position: anchor + offset, independently draggable.

Group 14 — Engine Toggle Button

Property	Default	What it does
Show Engine Toggle Button	true	Draws a clickable on-chart button that toggles this chart's engine on/off.
Show Panels Button	true	Draws the segmented **State · Summary · Table** button that shows/hides the three draggable panels. Green = visible, gray = hidden. The engine keeps running; lines/markers stay visible. Draggable. Renamed from *Show State Panel Toggle Button*. *(2026-07-09)*
Toggle Button Horizontal Anchor	Left	Measure button position from the left or right edge.
Toggle Button Vertical Anchor	Top	Measure button position from the top or bottom edge.
Toggle Button Offset X (%)	1.0	Horizontal distance from the anchored edge, as % of panel width. Draggable.
Toggle Button Offset Y (%)	2.0	Vertical distance from the anchored edge, as % of panel height. Draggable.

Property	Default	What it does
Button Opacity (%)	(group)	Background opacity of the three chart buttons (Panels, Auction Engine, Analyze). Button text stays fully opaque.

> **Per-chart toggle.** The engine button turns the engine on/off **only on the chart it sits on** — other charts and instruments are unaffected. When OFF, the engine keeps classifying in the background (history stays current) but hides its visuals and reports itself as "not present" to Suite consumers (VolDrive/VolImbalance fall back to their location-blind behavior). The button stays visible so you can switch the engine back on.

Group 16 — Auction Analyst

Property	Default	What it does
Show Analyst Button	false	Shows an "Analyze" button next to the engine toggle. Clicking it sends the current auction snapshot to the Flow.42 backend and shows a narrative analysis. Requires an active Flow.42 Trading Suite subscription for the saved license id.
Analysis Language	English	Language of the narrative analysis: German, English, Brazilian Portuguese, Spanish, Italian, French, or Simplified Chinese.
Window Text Color	Terminal Green	Text color of the result window: plain White, or the classic green-phosphor Terminal Green (green on black, CRT style).

11. Suite Integration

The AuctionEngine can share its auction context with Flow.42 Trading Suite indicators (VolDrive, VolImbalance) through a loose, late-bound bridge.

- **Optional & independent.** The AuctionEngine is a separate product. VolDrive/VolImbalance run normally whether or not the engine is installed. If the engine

is absent, not on the chart, on a different instrument/interval, or toggled off, the consumers simply skip the auction layer.

- **Tooltip context line.** When the engine is present, the consumer's existing tooltip gains one extra line: Auction Engine: — Continuation, Mean Reversion, Potential Reversal, or Neutral. No second tooltip window is created.
- **Optional visual modulation.** Consumers can dim conditions that run counter to the auction thesis. **VolDrive** defaults to *Subtle* dimming (it is a trend indicator). **VolImbalance** defaults to *Off* (reversal-oriented; the context appears in the tooltip but markers are not dimmed unless the trader opts in).

The bias logic: a read *with* the auction thesis is **Continuation** (full opacity). A counter read while the thesis is intact is **Mean Reversion** (slightly dimmed). A counter read after the thesis breaks is **Potential Reversal** (strongly dimmed). No dominant thesis → **Neutral** (no change).

Modulation modes

- **Live** — the engine's current classification, applied to every visible read every frame. Matches live trading; past reads are re-judged by the present state, so it is not a faithful record of the past. Default.
- **Historical** — the classification the engine recorded at each read's own bar, so a past read reflects the auction as it was at that moment. Lets you study behavior after the fact, including on a closed market. Falls back to no modulation for bars older than the engine's rolling history window (~2000 bars).

12. Troubleshooting

Symptom	Fix
Panel shows "Prior: load more history" or stays "Initializing."	The chart does not yet contain a full prior session. Load more historical bars so at least one complete prior session is included.

Symptom	Fix
Model line shows “Bar Distributed (Approximation).”	The profile is approximated, not built from true Volume-at-Price. Switch the data series to Tick Replay, or set the model to Volumetric Bars.
The engine does not affect VolDrive/VolImbalance.	Is the AuctionEngine on the same chart (same instrument + interval)? Is the toggle button ON? In VolImbalance, visual modulation is Off by default — only the tooltip line appears unless you enable it.
Lines extend into after-hours and clutter the chart.	Enable No After Hours Lines (Trim Lines After Session End).
State-change markers overlap other indicators at the top.	Increase Label Top Offset Pixels.
Panel color changes between two markers (gold on a “FAIL DOWN” stretch).	Expected — see §7. The panel shows the true per-bar state; markers only flag major transitions.
Toggling one chart’s engine changed another chart.	A defect in earlier builds (a global toggle). The current build is strictly per chart. Recompile all files to the current version.
AI Analyst window says “No Flow42 license id is configured.”	Enter your Flow42 license in the Product Information section, then click Analyze again.
AI Analyst window says “No auction data available yet.”	The profiles are still initializing. Wait until the panel shows a classified state, then analyze again.

13. Change Log

Dated record of user-visible changes to the AuctionEngine. Newest first.

Date	Change
2026-07-09	**Panels button.** The *State Panel* toggle became a segmented **State · Summary · Table** button, so the Summary box and the Overview table can be shown/hidden directly on the chart instead of only through the properties grid. Green = visible, gray = hidden.

Date	Change
	The property *Show State Panel Toggle Button* was renamed to *Show Panels Button*. Panel toggles are view-only and reset to the configured defaults when the chart reloads.

Flow.42 Analytics

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The AuctionEngine is a context and visualization tool and does not provide trade recommendations.

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